ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** a simplified representation of a system that keeps what matters for the question	Models leave out unimportant details.
2	**(a)** an approximate answer is enough for a decision	Orders of magnitude.
3	**(b)** a law describes what happens while a theory explains why	Different roles.
4	**(a)** a model	A visual model of matter.
5	**(a)** 10000 litres	0.5 x 20000.
6	**(d)** a general principle that applies in every process	A fundamental law.
7	**(a)** describes a regular pattern observed in nature	Laws describe regularities.
8	**(d)** refine the model	Science corrects itself.
9	**(b)** burning of magnesium	New substance formed.
10	**(c)** explains many observations and makes testable predictions	Theories explain and predict.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Simplified representation of a real system (1); Ball-and-stick model of a molecule (1).	1+1
12	It is a deliberate assumption to simplify (1); The model is accurate enough for the purpose (1).	1+1
13	Examine assumptions and measurements (1); Revise the model and test again (1).	1+1
14	Law describes a regular pattern (1); Theory explains why it happens (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Choose features important to the question (1); Ignore minor details such as air resistance for a falling ball (1); Assumptions make the problem manageable (1).	1+1+1
16	Units were confused (litres and gallons or pounds and kilograms) (1); Wrong quantity of fuel was loaded (1); Care with units is vital (1).	1+1+1
17	Mass = 2.5 kg (1); = 2500 g (1); Density used (1).	1+1+1
	OR: One breath about 0.5 litre (1); About 20000 breaths a day (1); 0.5 x 20000 = 10000 litres (1).	1+1+1
18	Treated matter in stars as ionised gas (1); Applied known laws of physics and chemistry (1); Explained the different spectra of stars (1).	1+1+1
	OR: Accuracy: closeness to the true value (1); Precision: closeness of repeated readings (1); Example: 15.2 and 15.3 cm are precise (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Question from observation (1); Hypothesis and prediction (1); Fair test experiment (1); Data collection and analysis (1); Conclusion and further questions (1).	1+1+1+1+1
	OR: Instrument limits (1); Human errors like parallax (1); Repeat measurements (1); Take the average (1); Use better instruments (1).	1+1+1+1+1
20	Ball and stick model (1); Atomic models (Dalton to Bohr) (1); Ideal gas or frictionless motion (1); New evidence leads to revision (1); Models are useful, not final (1).	1+1+1+1+1
	OR: Medicines (1); Fertilisers (1); Plastics (1); Cleaning products (1); Responsibility: safety and environment (1).	1+1+1+1+1

Mistakes to avoid:

- Multiplying by $18/5$ instead of $5/18$ (or the reverse) when converting between km/h and m/s; remember that m/s values are always *smaller* numbers.
- Writing a measured value without its unit, or mixing units (cm with m) in one calculation.
- Confusing *law* with *theory*: a law describes a pattern, a theory explains it; a theory does not 'become' a law.
- Taking the mean of the readings as 'the true value'; it is only a more reliable estimate, and a single reading is not 'wrong' just because it differs from others.
- Treating a model as an exact copy of reality, or saying that a model that fails a test should be ignored instead of being improved.
- Mixing up accuracy and precision, and reading the least count of an instrument wrongly (use the smallest division, not the numbered one).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** units must be used carefully and consistently	Importance of units.
2	**(b)** new evidence required changes	Models are revised.
3	**(a)** re-examine the model, assumptions or measurements	Mismatch leads to improvement.
4	**(d)** making an assumption to simplify the model	Idealisation.
5	**(c)** small differences grow in complex systems	Chaos in the atmosphere.
6	**(a)** everyone can compare and repeat measurements	Common standards.
7	**(d)** 'If I increase the length of a pendulum its time period will increase.'	Testable statement.
8	**(a)** 2500 g	1 kg = 1000 g.
9	**(a)** kilogram	Base unit of mass.
10	**(d)** the composition, properties and transformations of matter	Definition.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gives a quick approximate answer to check feasibility (1); Air breathed in a day is about 10000 litres (1).	1+1
12	To describe patterns precisely (1); To make predictions (1).	1+1
13	Everyone can compare and repeat measurements (1); Avoids confusion and mistakes (1).	1+1
14	Metre (1); Kilogram and second (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hypothesis: testable proposal (1); Law: describes a pattern (1); Theory: well-tested explanation (1).	1+1+1
16	Weather is a very complex system (1); Small differences in data grow with time (1); Models are only approximate (1).	1+1+1
17	Simplify the problem (1); Show the main behaviour (1); Can be corrected later (1).	1+1+1
	OR: To communicate without confusion (1); c for the speed of light (1); m for metre, kg for kilogram (1).	1+1+1
18	Make measurable observations like clouds, humidity (1); Note whether rain falls (1); Compare prediction with outcome (1).	1+1+1
	OR: Observation (1); Questioning and reasoning (1); Careful measurement and honest reporting (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Air breathed per day: about 10000 L (1); Number of heartbeats: 72 x 60 x 24 about 100000 (1); Estimation checks feasibility (1); Exact: medicine dose (1); Exact: fuel loading (1).	1+1+1+1+1
	OR: Papers and presentations (1); Others repeat experiments (1); Mistakes are corrected (1); Fake data harms everyone (1); Trust in science (1).	1+1+1+1+1
20	SI base units: m, kg, s (1); Derived units: m/s, N (1); Prefixes: kilo, centi, milli (1); 1 km = 1000 m, 1 m = 100 cm (1); 1 m = 1000 mm (1).	1+1+1+1+1
	OR: Drop balls of different mass from same height (1); Measure fall times (1); Heavy and light balls reach almost together (1); Air resistance affects light, large-area objects (1); Prediction is refined (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing a measured value without its unit, or mixing units (cm with m) in one calculation.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3000 m	1 km = 1000 m.
2	**(a)** metre	Base unit.
3	**(b)** ionised gas obeying the laws of physics and chemistry	Ionisation equation.
4	**(c)** 7200 s	2 x 3600.
5	**(b)** second	Base unit.
6	**(b)** describing patterns precisely and making predictions	It is the language of patterns.
7	**(b)** 10 m/s	36 x 1000/3600.
8	**(d)** scientists agree on common symbols to communicate clearly	Standard symbols.
9	**(c)** measurements have some uncertainty	Small variations are normal.
10	**(c)** the dose of a medicine	Exactness is vital.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Theories are improved with new evidence (1); Science corrects itself (1).	1+1
12	Statement of what should happen based on a model, which can be tested (1); A guess has no reasoning or test (1).	1+1
13	5 x 1000 / 3600 (1); = 1.39 m/s (1).	1+1
14	Study of matter, its properties and changes (1); Medicine and materials (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A theory is supported by much evidence (1); It explains many observations (1); It makes predictions that can be tested (1).	1+1+1
16	Avoids false conclusions (1); Allows testing of ideas (1); Helps others to repeat the work (1).	1+1+1
17	Day: 24 x 3600 = 86400 s (1); Year: 86400 x 365 (1); About 3.15 x 10^7 s (1).	1+1+1
	OR: (a) 72 x 5/18 = 20 m/s (1); (b) 5 x 18/5 = 18 km/h (1); Use 1 km = 1000 m and 1 h = 3600 s (1).	1+1+1
18	Burning: heat and light (1); Rusting: brown solid (1); Souring of milk: sour taste (1).	1+1+1
	OR: Medicines (1); Fertilisers and food preservation (1); Materials like plastics and cement (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hypothesis: tentative explanation (1); Law: Newton's law or conservation of energy (1); Theory: atomic theory (1); Testing: predictions compared with experiments (1); Revision if they fail (1).	1+1+1+1+1
	OR: Wrong units caused a shortage of fuel (1); 1 km/h = 5/18 m/s (1); 1 L = 1000 mL (1); 1 kg = 1000 g (1); Always check units (1).	1+1+1+1+1
20	Saha ionisation equation (1); Explained stellar spectra (1); Used mathematics with physics and chemistry (1); Predictions about stars (1); Mathematics describes patterns (1).	1+1+1+1+1
	OR: Models use physics laws (1); Data from satellites (1); Predictions for a few days are good (1); Longer forecasts less reliable (1); Models are improved by data (1).	1+1+1+1+1

Mistakes to avoid:

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